Rashelle Belokon
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Final Project Report

The Impact of Breast Density and Other Risk Factors on Breast Cancer Risk

Introduction

Breast cancer will impact one in eight American women at some point in their life. In 2020, there were 2.3 million women diagnosed with breast cancer and 685,000 deaths globally caused by it. In the current day and age, we have a lot of advancements that make breast cancer less deadly than ever before, and detection methods which allow us to spot the risk before it becomes dangerous. Most notably, mammograms are used. This is a method widely known and accepted, and yet one factor can impact the effectiveness: breast density. There are four categories of breast density: fatty, scattered, heterogeneously dense, and extremely dense. Women who have been told they have dense breasts should be aware that it can make it difficult for screening mammography to detect tumors.

Women who have dense breasts can also be at increased risk for breast cancer. With conventional mammography, while we can be as accurate as 98% in a fatty breast, our sensitivity can drop to as low as 30% in women with extremely dense breasts. Having extremely dense breasts can make you 2 times more likely to develop breast cancer in comparison with those who have scattered breast density, and 1.5 times more likely for heterogeneously dense breasts. Since the only way to test for breast density currently is a mammogram, not being aware of the chance of having high breast density may delay diagnosis of breast cancer when it is potentially more likely to occur. This may make treatment options more limited and complicated to administer.

Data Sources and Experimental Design

This project aims to focus on predicting breast cancer risk with a highlighted focus on breast density as a less discussed risk factor compared to some others like age and BMI. It builds off of a previous study I performed on this data, in which I examined how breast density was correlated with other risk factors (age, BMI, and HRT) and how breast cancer and false negative mammogram results were correlated with breast density using simple bar charts. I am using a risk factor dataset obtained through the BCSC Organization which organizes risk factor data for approximately 2 million records. With this data, I aim to use more sophisticated ML techniques to expand on these relationships in a more interpretable way, namely by creating predictive models that estimate breast cancer risk off of a variety of factors. The variables and their summary statistics are presented in Table 1 below.

Table 1. Summary Statistics for Variables

Variable	Mean	Standard Deviation	Minimum	Maximum
Breast Density	2.448	0.743	1.0	4.0
Age Group	57.240	11.761	18.0	85.0
BMI Group	21.842	9.577	10.0	35.0
Previous breast biopsy	0.252	0.434	0.0	1.0
Breast Cancer History	0.074	0.261	0.0	1.0
Race/Ethnicity	1.770	1.276	1.0	6.0
Family History	0.181	0.385	0.0	1.0
Age Menarche	0.900	0.673	0.0	2.0
Age First Birth	2.011	1.304	0.0	4.0
Current HRT	0.0153	0.123	0.0	1.0
Menopause	1.775	0.552	1.0	3.0

I implemented regularization in logistic regression to perform model selection on a wide range of available risk factors: age, race/ethnicity, family history, age at menarche, age at first birth, HRT usage, menopausal status, BMI. I evaluated my resulting model with a binary cross-entropy approach, with a goal of creating a model that accurately estimates cancer risk. This model was built from scratch and compared to a Sklearn-built model. I used a similar method with a random forest approach to allow for ranking feature importance, which I evaluated on specificity and sensitivity results.

For validation, I employed stratified sampling for 10-fold cross validation, stratified in order to account for class imbalance in cancer diagnosis as well as various categoricals such as race and breast density. The regularization in my logistic model also aims to address any issues of multicollinearity between risk factors.

In terms of data preprocessing, this data is very organized and having worked with it before I have a good background in it. In my past work with this data, as I was studying false negatives by breast density but the nature of how screenings were done didn't specify false negative results, I instead considered a datapoint a false negative if the mammogram assessment was negative and cancer was diagnosed within a year of the mammogram, and may employ similar logic in the models I create here. For risk factor analysis here, I used past incidence of breast cancer as the variable of interest to get a sense of aggregate risk over an individual's lifespan.

There are some ethical considerations to keep in mind when studying such a topic. One key factor to consider is how players in the medical industry will use this information – namely, insurance companies. While understanding the risk factors that play a role in cancer risk is vital for early detection and has the potential to help many who are at-risk, it can also have the

adverse effect of hiding necessary procedures behind paywalls for many without insurance plans to cover those procedures. These companies base pricing and policy decisions on studies like this one, so it is very important to consider harm reduction when looking into these topics.

Results and Performance Metrics

I found that breast density was a significant factor in breast cancer risk, along with being strongly correlated with other risk factors such as BMI and age group. BMI displayed the most significant relationship with Breast Density, while outside of age, prior history of breast biopsy or aspiration had the highest feature importance, BMI and Breast Density being right behind.

Logistic Regression

I employed Sklearn logistic regression along with ridge penalty to lessen the impact from less important variables. I achieved a model which misses many positives and suggests some false positives (Exhibit A). This model gives a good idea of probabilities of developing breast cancer at some point in life. It's imperfect – namely due to the age factor being a poor predictive factor since the variable of interest was 'history of cancer', meaning higher age groups will always be more likely to have such history. Nevertheless, it gives a good idea of risk factors interactions with breast cancer risk.

I compared these results to ones derived by gradient descent employing similar logic. The resulting model was much more lenient with its positive labeling, ending up catching a large majority of true positives (16399 out of 18994), but also flagging many true negatives as positives (75414) (Exhibit B). A majority of flagged positives were false positives, but for a medical problem of the like we are studying here, this would be the preferred outcome.

Random Forest

I employed a Random Forest approach to get feature importances. I made a model of factors on breast cancer history, and one of other risk factors on breast density itself (Exhibit C). This model produced a 92% accuracy with only 7% recall and 32% precision, similar results to the Sklearn logistic regression model, but giving us useful insights into feature importance. For cancer history, breast density was found to be in the top along with BMI group and age as highly important features for positive prediction. Age makes sense because of the aforementioned issue with the 'history of breast cancer' variable of interest, but breast density and BMI are useful indicators along with history of breast biopsy or aspiration.

I also used a Random Forest approach to see how the risk factor variables correlate with one another, and found age and BMI to be highly correlated with breast density (Exhibit D). Though not directly tied to breast cancer risk, the correlation between risk factors could be important to consider in interpreting previous results as the level of impact assigned to each variable could be muddled by relationships between them.

Conclusion and Future Opportunities

This study has created a model to catch the vast majority of high-risk individuals using risk factors such as BMI, history of operation, and breast density. It has also determined these to be variables of interest in predicting breast cancer risk. This can allow for medical practice to be enforced of more thorough investigative processes for high-risk individuals, for instance recommending MRI or other alternative testing measures for high-breast-density patients at high

risk. It can also spread the word to the masses who may not know about the potential risks they face in a medical world which too often ignores women and AFAB people.

One major limitation of this study was the variable of interest. Due to a lack of availability of comprehensive time series data on individuals in the database, it was difficult to track when cancer occurred to truly understand the risk factors relevant to the patient at onset, which especially made the age variable not useful to statistical analysis. There is definitely a future opportunity here to utilize a time-series approach, as well as to study the effectiveness of mammogram testing through a false negatives lens.

Bibliography

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Link to Data

- BCSC Risk Factor Dataset: https://www.bcsc-research.org/index.php/datasets/rf

Link to Github Repository for Replication

- Github: https://github.com/belokonr/Breast-Cancer-Study/tree/main

Appendix

Exhibit A

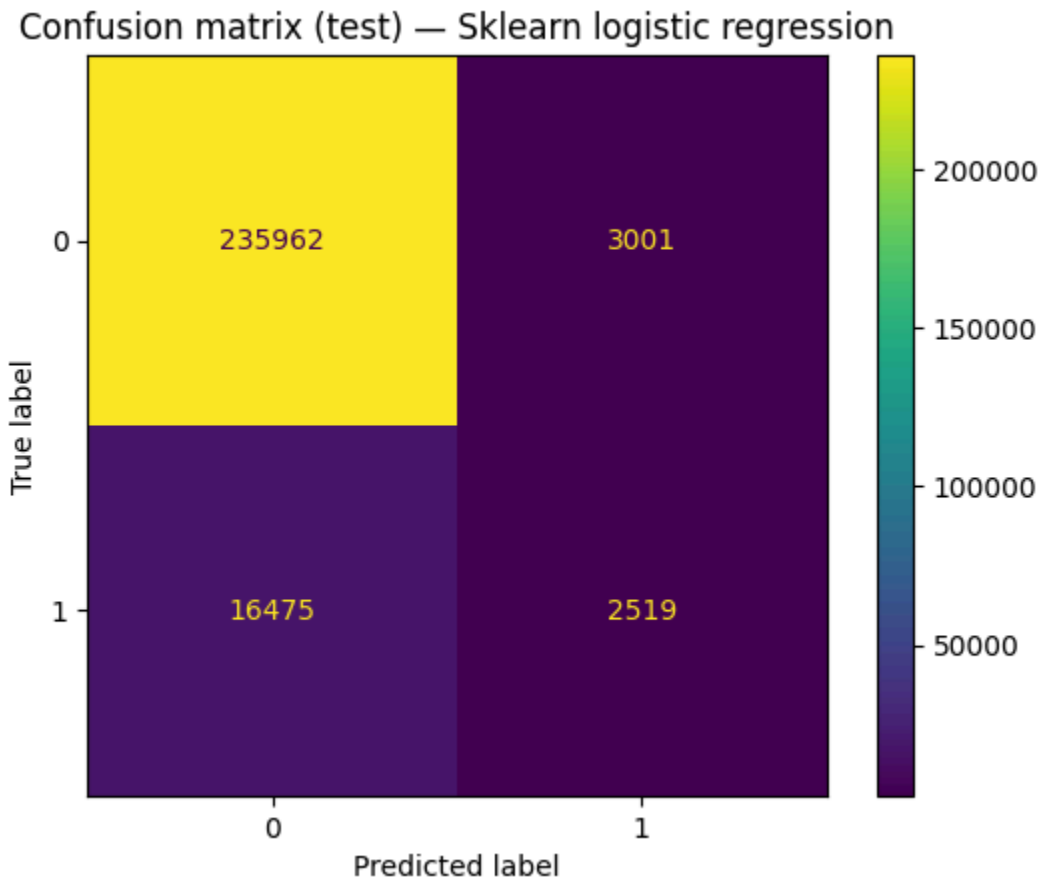
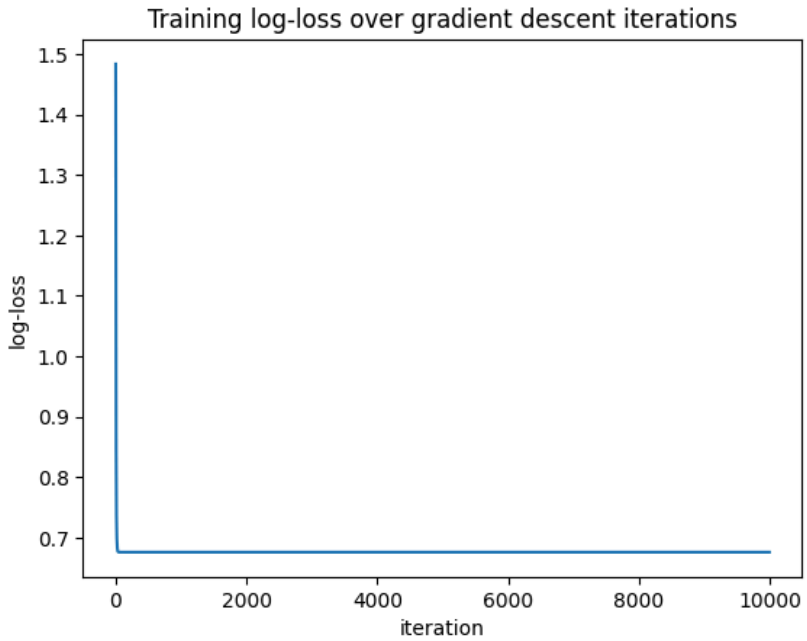
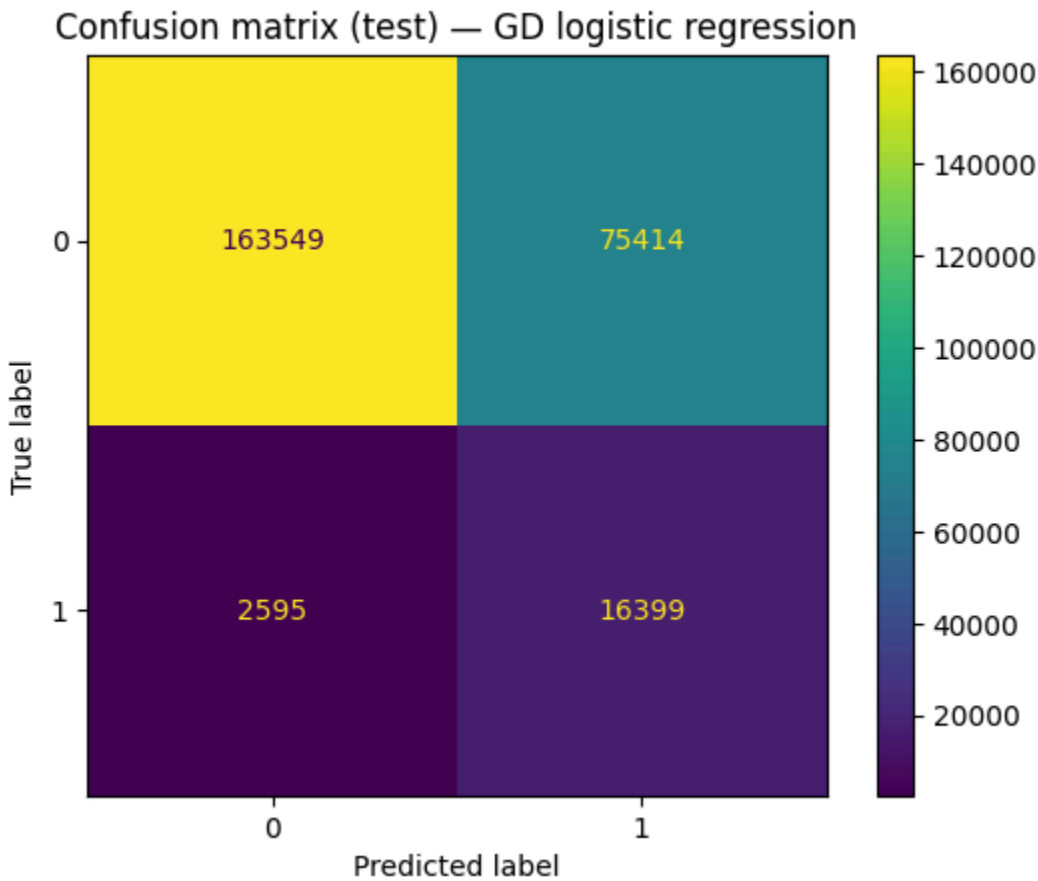

Confusion matrix (test) — Sklearn logistic regression
235962
3001
16475
2519
True label
Predicted label
0
1
0
1
200000
150000
100000
50000

Exhibit B


Training log-loss over gradient descent iterations
log-loss
1.5
1.4
1.3
1.2
1.1
1.0
0.9
0.8
0.7
0
2000
4000
6000
8000
10000
iteration


Confusion matrix (test) — GD logistic regression
True label
Predicted label
0
1
0
1
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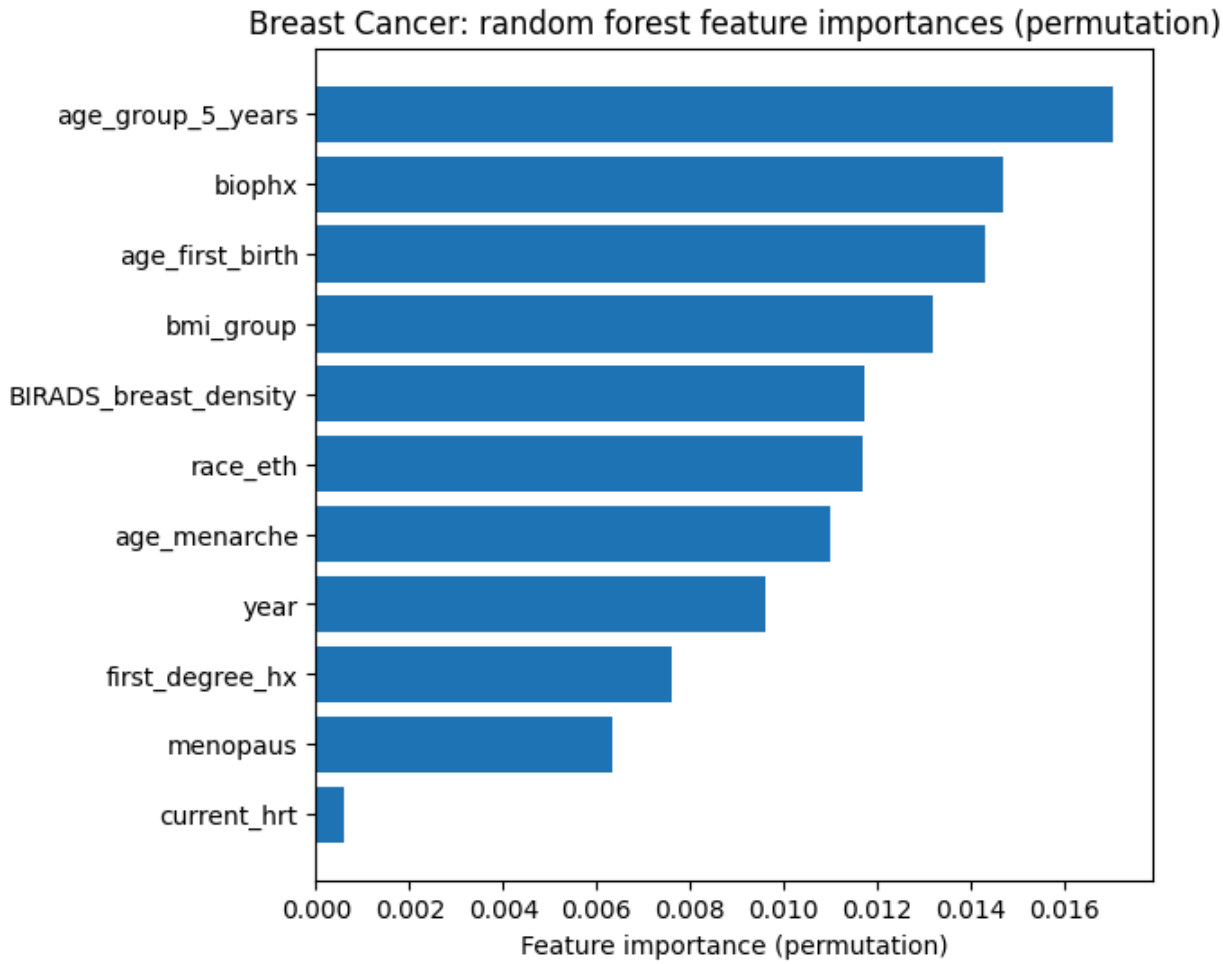
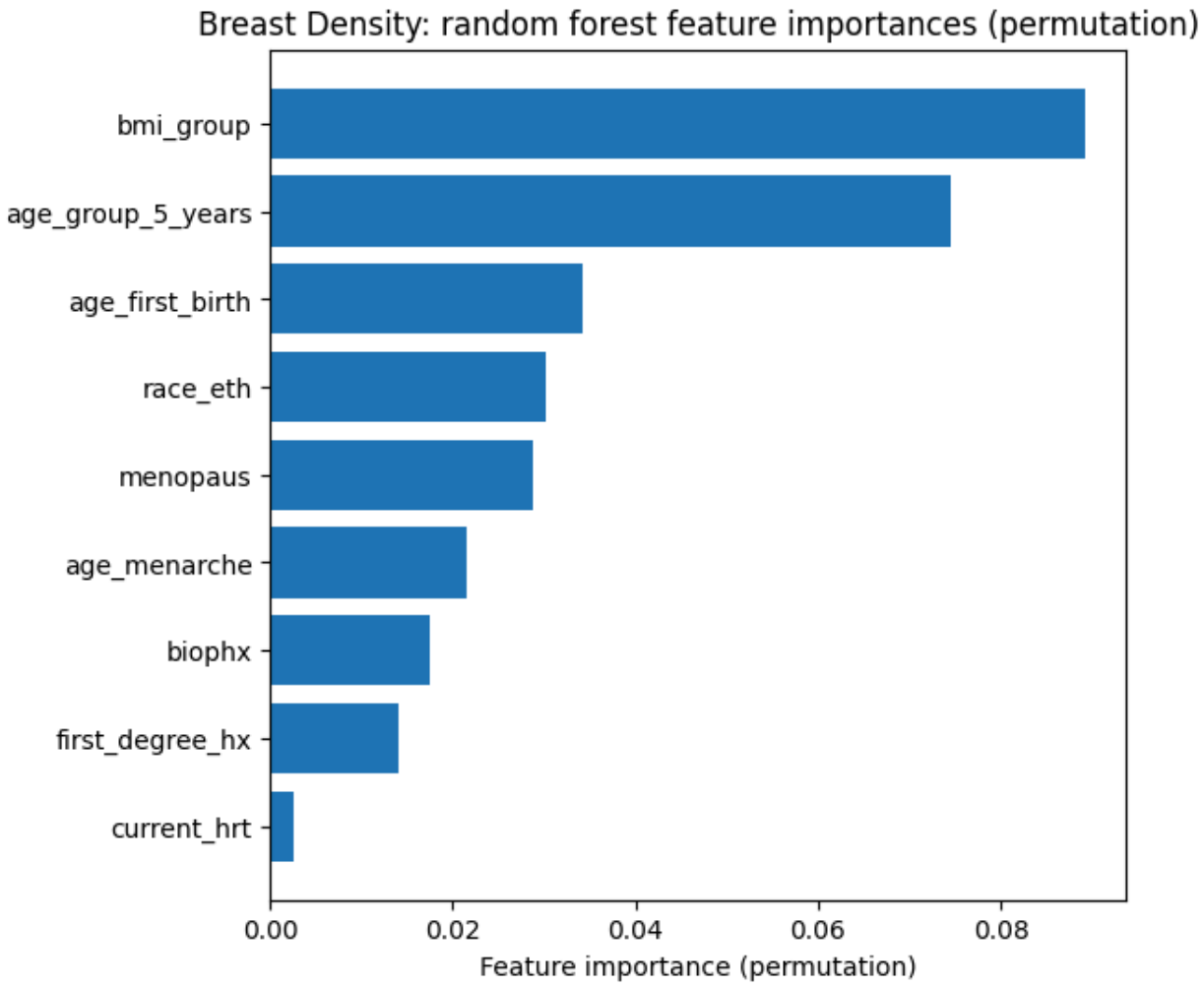
75414
2595
16399
160000
140000
120000
100000
80000
60000
40000
20000

Exhibit C



Metric	Score
Precision	0.3285
Recall	0.0768
Accuracy	0.9207
F1 Score	0.1245

Exhibit D


Breast Density: random forest feature importances (permutation)
bmi_group
age_group_5_years
age_first_birth
race_eth
menopaus
age_menarche
biophx
first_degree_hx
current_hrt
0.00
0.02
0.04
0.06
0.08
Feature importance (permutation)